# Lab 6 - Advanced Machine Learning

## Inference on Graphical Models

Ludovic TRAUTMANN

CentraleSupélec

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# General Informations

- **Assignment:** You will work **in pairs**.
- **Due date: Each assignment must be sent within 7 days of the lab session (by the next Thursday evening).**
  Send your notebook to `ludovic.trautmann@inria.fr` (only one submission per pair, remember to put both of your names on it).
- **Grading:** There are 6 lab sessions. The average grade will count for one third of your final grade.
  One part of the final exam will cover the material from the labs.

# Lesson Recap

$$\mathbf{X} \sim \mathcal{N}(\mu, \Lambda^{-1}), \quad p_X(\mathbf{x}) = \frac{(\det \Lambda)^{\frac{1}{2}}}{(2\pi)^{\frac{p}{2}}} \exp\left(-\frac{1}{2}(\mathbf{x}-\mu)^\top \Lambda (\mathbf{x}-\mu)\right) \tag{1}$$

The Maximum-Likelihood estimator of $\mu$ is $\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$

**Maximum Likelihood Estimator of $\Lambda$**

$$\underset{\Lambda \succ 0}{\text{minimize}} \; -\log\det\Lambda + \text{tr}(\mathbf{S}\Lambda), \quad \mathbf{S} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{\mu})(x_i - \hat{\mu})^\top \tag{2}$$

Solution :

$$\hat{\Lambda} = \mathbf{S}^{-1} \tag{3}$$

Unstable inverse $\rightarrow$ Reduced via a sparsity constraint

## Graphical LASSO : Problem formulation

$$\underset{\Lambda \succ 0}{\text{minimize}} \; -\log\det\Lambda + \text{tr}(\mathbf{S}\Lambda) + \lambda\|\Lambda\|_1 \tag{4}$$

$$\Downarrow \text{ (Variable splitting)}$$

$$\underset{\Lambda \succ 0, \mathbf{Z}=\mathbf{Z}^\top}{\text{minimize}} \; -\log\det\Lambda + \text{tr}(\mathbf{S}\Lambda) + \lambda\|\mathbf{Z}\|_1 \quad \text{such that} \quad \Lambda = \mathbf{Z} \tag{5}$$

$$\Downarrow \text{ (Lagrangian Formulation)}$$

$$\mathcal{L}_\rho(\Lambda, \mathbf{Z}; \mathbf{U}) = -\log\det\Lambda + \text{tr}(\mathbf{S}\Lambda) + \lambda\|\mathbf{Z}\|_1 + \frac{\rho}{2}\|\Lambda - \mathbf{Z} + \mathbf{U}\|_F^2 - \frac{\rho}{2}\|\mathbf{U}\|_F^2 \tag{6}$$

- $\Lambda$ and $\mathbf{Z}$ are the primal variables, with the condition $\Lambda = \mathbf{Z}$
- $\mathbf{U}$ is the scaled dual variable, associated with the constraint $\Lambda = \mathbf{Z}$
- $\lambda > 0$ controls the sparsity of $\Lambda$
- $\rho > 0$ controls how much we penalize the violation of the constraint

**Algorithm** ADMM for GLASSO

**for** $k = 1, \ldots$ **do**

$\mathbf{A}_{k+1} \leftarrow \rho(\mathbf{Z}_k - \mathbf{U}_k) - \mathbf{S}$

Diagonalize $\mathbf{A}_{k+1}$ in $\mathbf{A}_{k+1} = \mathbf{Q}_{k+1}\,\mathrm{Diag}(d_i^{k+1})\mathbf{Q}_{k+1}^\top$

$$\Lambda_{k+1} \leftarrow \mathbf{Q}_{k+1}\,\mathrm{Diag}\left(\frac{d_i^{k+1} + \sqrt{(d_i^{k+1})^2 + 4\rho}}{2\rho}\right)\mathbf{Q}_{k+1}^\top$$

$\mathbf{Z}_{k+1} \leftarrow \mathrm{SoftThreshold}_{\frac{\lambda}{\rho}}(\Lambda_{k+1} + \mathbf{U}_k)$

$\mathbf{U}_{k+1} \leftarrow \mathbf{U}_k + \Lambda_{k+1} - \mathbf{Z}_{k+1}$

**end for**

# Nodewise Regression

## Nodewise Regression

Learn the support of the precision matrix $\Lambda$ via sparse linear regressions

1. With $\mathbf{X} = (X^{(1)}, \dots, X^{(p)}) \in \mathbb{R}^p \sim \mathcal{N}(0, \Lambda^{-1})$, regress each dimension onto the others

$$\forall j = 1, \dots, p \quad X^{(j)} = \sum_{i \neq j} \hat{\beta}_i^{(j)} X^{(i)} + \epsilon^{(j)} \tag{7}$$

2. Use the linear regression coefficient as edge activators

$$\{i, j\} \in \mathcal{E} \Leftrightarrow \Lambda_{i,j} \neq 0 \Leftrightarrow \hat{\beta}_i^{(j)} \neq 0 \underset{\text{(also try OR)}}{\text{AND}} \hat{\beta}_j^{(i)} \neq 0 \tag{8}$$